

DataProphet

Demystifying Supervision Data Generalization in Multimodal LMs

Can you predict — *before any training* — how much a given supervision dataset will help a given target benchmark?

ICLR 2026 · code & data released (MIT)

What this deck covers

- **The question** — predicting data influence with zero training
- **What intuition gets wrong** — three counter-intuitive findings from a measured 14×14 matrix
- **The metric** — perplexity $\times$ similarity $\times$ diversity, and why it is a product
- **Results** — correlation, ablation, and data selection on real and synthetic pools
- **Assessment** — the one claim to reject, and the constraints to write down

The Question

Can influence be predicted before any training takes place?

The paper

Title	Demystifying Supervision Data Generalization in Multimodal LMs
Venue	ICLR 2026
Authors	Xuan Qi (UPenn / Tsinghua), Luxi He (Princeton), Dan Roth (UPenn), Xingyu Fu (UPenn / Princeton)
Code	<code>github.com/DataProphet26/dataprophet</code> — MIT, 17 files
Data	HuggingFace <code>THUQiXuan/DataProphet</code>
Project	<code>dataprophet26.github.io</code>

Completed during Xuan Qi's summer research internship at UPenn. Reproducibility is unusually good for an ICLR paper.

The question, stated precisely

Given a training dataset D_i , can we predict its influence on a target benchmark T_j **before any training takes place**?

- Influence is a **relative performance change**:

$$\Delta_{s \rightarrow t} = \frac{A_t^s - A_t}{A_t}$$

- i.e. how much target t improves after fine-tuning on source s ; the reverse is $\Delta_{t \rightarrow s}$
- The conventional approach trusts **intuitive task similarity** — text-rich vs vision-centric

The setup: a complete 14×14 matrix

Design

- Base model **InternVL3-2B**
- **20K** train / **1K** test per source
- **Fixed compute** — same volume, model, optimizer, scheduler, epochs, batch size
- Full-parameter SFT, 1 epoch, lr $1e-5$ (LLaMA-Factory)

14 datasets, 7 task families (2 each)

- **OCR** — OCR-VQA, ScreenQA
- **Chart** — ChartQA, Chart2Text
- **Document** — DocVQA, FinanceQA
- **General VQA** — A-OKVQA, VC-GVQA
- **Spatial** — Open-Spatial, CLEVR-Relation
- **Counting** — CLEVR-Counting, TallyQA
- **Map** — GeomVerse, MapQA

Every claim in the paper rests on this one matrix — measured, not assumed.

What Intuition Gets Wrong

Three findings from the measured matrix

Finding 1 — influence is *asymmetric*

Open-Spatial → DocVQA

15.92%

DocVQA → Open-Spatial

26.14%

- $\Delta_{s \rightarrow t} \neq \Delta_{t \rightarrow s}$ — influence is not a symmetric similarity score
- Diagonal gains are still largest: same-distribution data remains most influential

Implication: influence is a **directed graph**, not a scalar property of a dataset. Any single "how good is this dataset" number discards this by construction.

Finding 2 — same task family $\neq$ most help

After training on **OCR-VQA**:

ScreenQA — also OCR

+17.88%

GeomVerse — map understanding

+21.74%

Training on OCR helps a *map* task more than it helps another *OCR* task. The paper's phrasing: influence is decided by the **individual dataset**, not the task category.

Finding 3 — text-rich helps vision-centric most

The intuition: OCR and charts both require extracting text and numbers from images, so OCR data should help chart tasks most.

After training on **ScreenQA** (an OCR dataset):

Text-rich targets

- ChartQA — +5.61%
- Chart2Text — +6.13%

Vision-centric targets

- Open-Spatial — +12.86%
- CLEVR-Relation — +7.96%

The result is the **reverse** of the intuition. Surface similarity does not predict transfer.

**Intuition about task similarity
is an unreliable guide
to transfer**

The Metric

A predictor you can actually compute

DATAPROPHET, in one equation

$$\mathcal{M}(s \rightarrow t) = \text{QSim} \cdot \text{ASim} \cdot \text{ISim} \cdot \text{PPL}(s) \cdot \frac{\text{Sil} + H}{\text{PPL}(t)}$$

- **QSim** / **ASim** / **ISim** — question, answer and image similarity between source and target
- **PPL(s)** / **PPL(t)** — multimodal perplexity of the source, divided by the target's
- **Sil + H** — source question diversity (silhouette + cluster-balance entropy)
- Every component is computed with the **frozen base model** — no training anywhere

Component 1 — multimodal perplexity

$$\text{PPL}(\mathcal{D}) = \exp(-\mathbb{E}_{\mathcal{D}} \mathbb{E}_t \log p_{\theta}(a_t \mid I, \tau_Q(Q), a_{<t}))$$

- **Hypothesis:** intrinsically *harder* source data provides greater value for enhancing capability
- Precedent in LLM work — perplexity as a data-quality estimator for corpus pruning
- **Insufficient alone:** perplexity-only gives $\tau_{\text{Tgt}} = 0.274$

It is, however, the single most important factor once combined — see the ablation.

Component 2 — cross-dataset similarity

- If source data resembles the target in the model's **embedding space**, it should be closer to "learning the target distribution"
- Encode **questions, answers and images separately** with the base MLLM's **frozen encoders**, ℓ_2 -normalize, take expected cosine similarity → **QSim, ASim, ISim**

A negative result worth recording: encoding question and answer *jointly* as **QASim** was **6% worse** in τ than keeping them separate.

Still not enough on its own: $\text{PPL} \times \text{similarity}$ gives $\tau_{\text{Tgt}} = 0.658$.

Component 3 — source question diversity

- **Hypothesis:** a source whose questions cover more ground encourages more generalizable skills
- Embed each source question, **K-means into K = 10** clusters, then measure separation and balance

Silhouette — separation

$$\text{Sil} = \mathbb{E}_{u \sim U} \left[\frac{b(u) - a(u)}{\max\{a(u), b(u)\}} \right]$$

$a(u)$ = mean intra-cluster distance; $b(u)$ = nearest other cluster.

Entropy — balance

$$H = -(\log K)^{-1} \sum_{k=1}^K \pi_k \log \pi_k$$

π_k = empirical cluster proportion.

Diversity = **Sil** + **H**: larger means broader coverage with better-separated, better-balanced clusters.

Why a product — and how it is scored

Successful transfer requires **simultaneous** alignment in text, vision, difficulty relative to the base model, and question coverage. The product form therefore **down-weights the score when any single factor is weak**.

- A weighted **sum** would let a strong factor mask a weak one — a product does not
- **Two-way protocol**, because influence is asymmetric: $\tau = (\tau_{\text{Tgt}} + \tau_{\text{Src}})/2$
 - τ_{Tgt} — fix a target, rank the 14 sources, compare with reality
 - τ_{Src} — fix a source, rank the 14 targets, compare with reality

Results

Prediction, ablation, and selection

Does it predict the rankings?

τ_{Tgt} average

0.863

range 0.736 – 0.933

τ_{Src} average

0.857

range 0.822 – 0.911

Headline

86.0%

Kendall's τ , both directions

- Consistently high across all 7 task families — not driven by one easy target
- The metric captures the **asymmetry**, which a symmetric similarity score could not

Ablation: what actually carries the signal

Variant	Kendall's τ (avg)
Full DATAPROPHET	0.860
w/o Answer Similarity	0.810
w/o Question Similarity	0.778
w/o Diversity (Silhouette & Entropy)	0.659
w/o Image Similarity	0.625
w/o Perplexity	0.487

Ordering: **perplexity** > **image similarity** $\approx$ **diversity** > **text similarity**.

The heuristics that did *not* work

All of these were tried, and none improved prediction:

- Question difficulty
- Model's familiarity with the image
- Model's familiarity with the question
- Answer length — adding it **decreased τ by 0.15**

Most "sensible-looking" data properties do not predict transfer. Publishing this list is more useful to successors than publishing only what worked.

From prediction to selection

Fixed budget $N = 280K$ — reweight each source by its DATAPROPHET score.

Real data

Method	Avg	Δ
Uniform	67.6	—
ICONS	69.6	+2.0
Oracle	70.8	+3.2
D.P.	71.0	+3.4

Synthetic data (~1.4M pool)

Method	Avg	Δ
Uniform	55.1	—
ICONS	60.8	+5.7
D.P.	62.0	+6.9

D.P. beats the trained selector in both settings — reportedly **edging the Oracle by 0.2%**.

It also works for RL, and ranks the generators

- **GRPO** via **verl** on Qwen2.5VL-3B-Instruct, prompt budget 300×14:
 - Real RL data: no-RL 55.7 → Equal 58.3 → **D.P. 59.5**
 - Synthetic RL data: baseline 56.4 → Random 56.4 → **D.P. 57.7**
- So the SFT conclusion **transfers to post-training** — a genuine broadening

A neat by-product: of the selected synthetic items, **38% came from GPT-5** and **62% from Gemini 2.5 Pro** — the selector implicitly ranks Gemini's data higher.

Assessment

What to take, and what to discount

What to take

- **Treat influence as a directed quantity.** $\Delta_{s \rightarrow t} \neq \Delta_{t \rightarrow s}$ is measured, so a single quality score per dataset is structurally insufficient. **This is the big one.**
- **"Looks similar" is not evidence.** If mixture weights come from task-similarity intuition, this paper says that heuristic is unreliable
- **Perplexity is the best single cheap signal** — and it points the same way as the dedup finding that high-frequency repetition means low information density
- **Multiply when conditions must hold jointly** — sums hide weak factors
- **Publish the failed heuristics** — four of them here, all plausible, none predictive

What to discount

- **"Beats the Oracle by 0.2%" is not a real result.** No seeds, no variance, no confidence intervals — 0.2% on a 14-benchmark macro average is indistinguishable from noise. Worse, the Oracle is built from **20K-per-source** observations extrapolated to a 280K mixture, exactly as D.P. is, so it is not a strict upper bound
- **Everything rests on one 2B model** (InternVL3-2B). The matrix, the ablations, the gains — no second family, no larger model
- **"Training-free" is not "model-free."** PPL uses the base model and the similarities use its frozen encoders, so **scores do not transfer across models** — change the base and recompute everything
- **2 datasets per task family** is thin support for a general claim about task *categories*
- **Small scale:** 20K samples, 1 epoch. Real instruction tuning is 100K–millions, multi-epoch
- **The Gemini-vs-GPT-5 comparison is confounded** — the two generators saw *different* image halves

Takeaways for our own work

- **Model data value as a matrix, not a vector.** If you must collapse it, be explicit about which direction you are collapsing
- **Budget for a cheap scorer.** Perplexity over candidates plus one embedding pass is far cheaper than a proxy sweep — but not free at millions of candidates
- **Write down the base-model dependency.** Any embedding- or perplexity-based score is bound to the model that produced it; that must be stated before anyone plans to reuse it
- **Use a selector's preferences to rank sources** — "62% of picked items came from Gemini" is a lightweight, useful way to compare data generators
- **Reproducibility is achievable** — MIT code, public dataset, project page. Worth matching

Questions

Full written analysis (Chinese + English) lives in `my-wiki/05-References/` .

Thank you